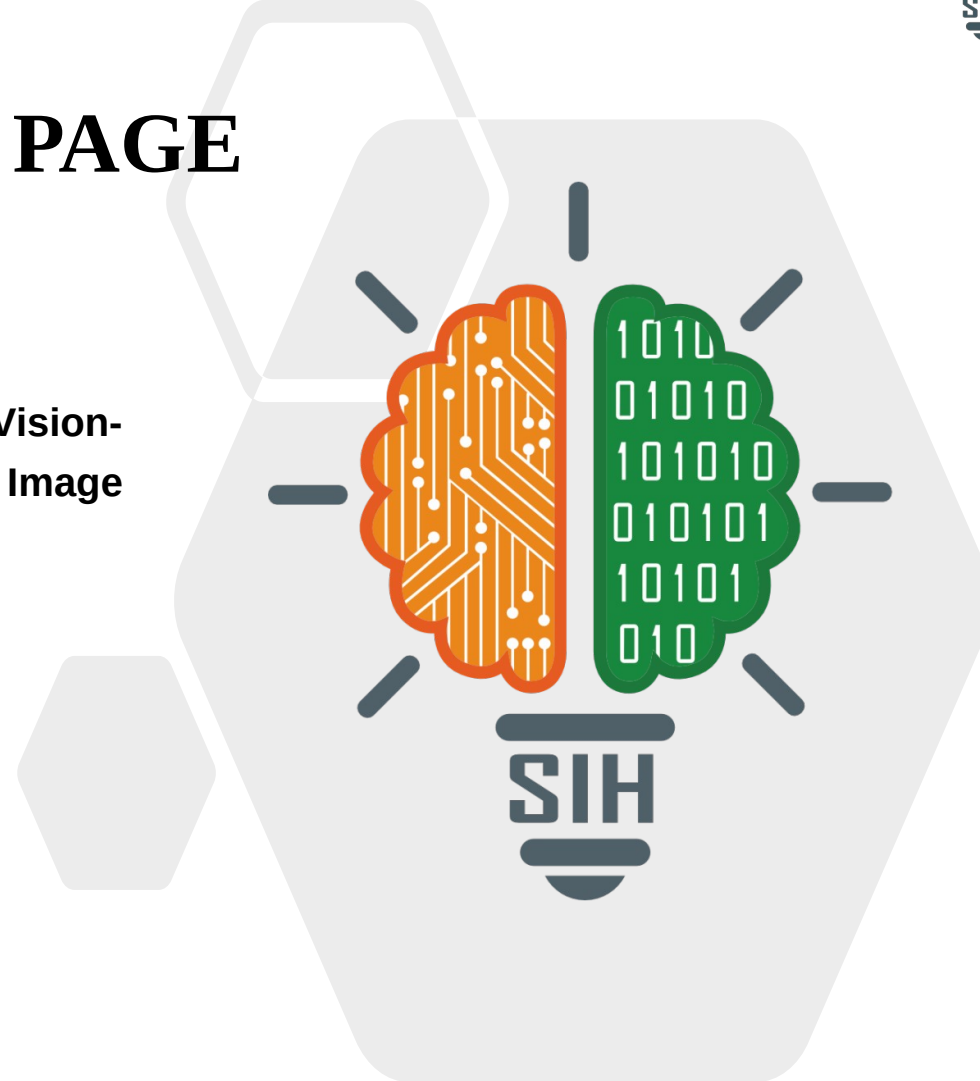




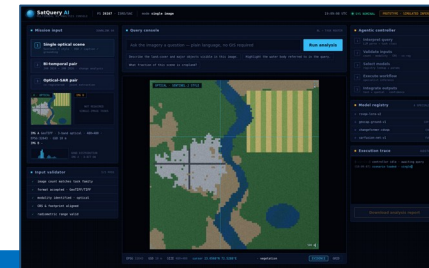
TITLE PAGE

- Problem Statement ID – 26167
- Problem Statement Title- SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- Theme- Space Technology
- PS Category- Software
- Team ID- (as registered on portal)
- Team Name (Registered on portal)- Lunar Circle

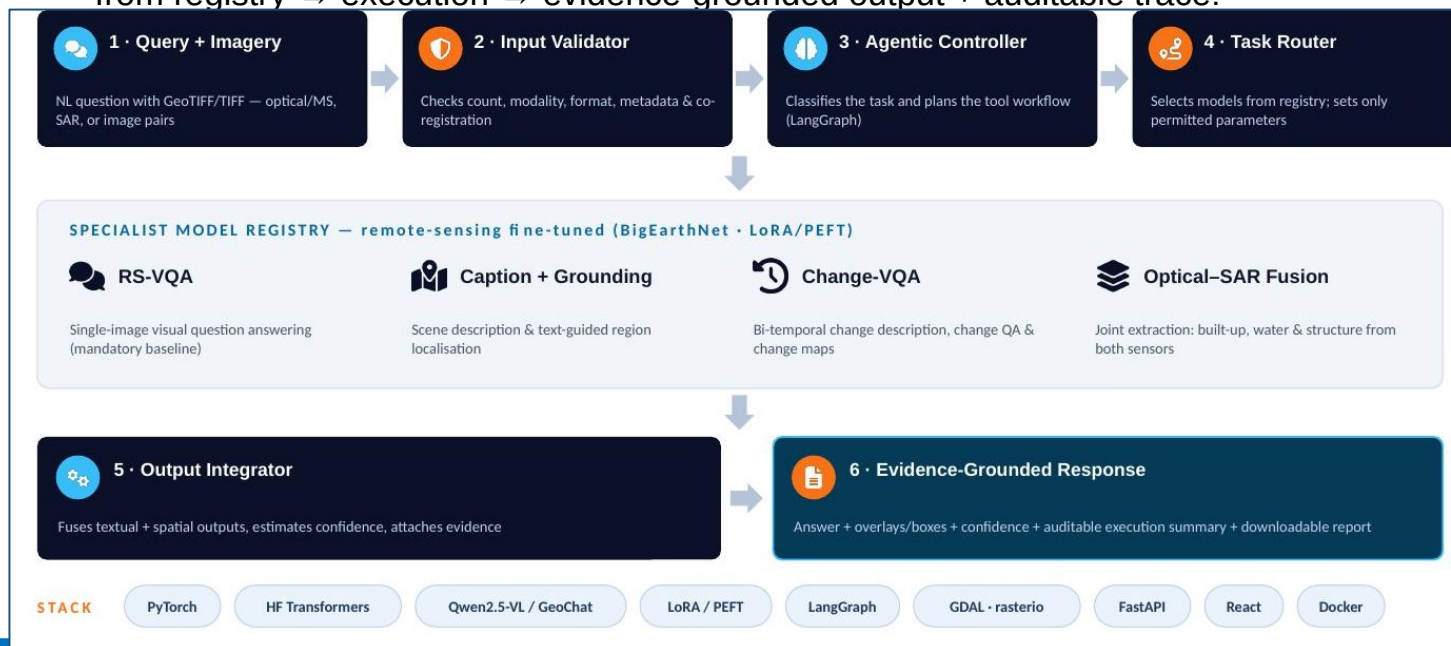


❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **What it is:** An agentic vision-language assistant: upload GeoTIFF imagery, ask in natural language, get an evidence-grounded answer (masks/boxes + confidence + auditable trace).
- Supports all 3 prescribed inputs — single optical/SAR image, co-registered optical–SAR pair, bi-temporal pair — across VQA, captioning, grounding, change-VQA and fusion.
- **How it addresses the problem:** Replaces fragmented single-task RS tools; a controller validates inputs and routes each query to fine-tuned specialist models — no GIS expertise needed.
- **Innovation & uniqueness:** Agentic orchestration (not one generic VLM); LoRA adaptation on BigEarthNet.txt — 464,044 co-registered S1 SAR + S2 pairs, 9.6M annotations; rejects incompatible inputs; ships visual evidence + execution summary.
- **LIVE:** Working prototype already deployed — gangadhar017.github.io/Lunar_circle (scan QR →)



- **Technologies:** Python · PyTorch · HF Transformers · Qwen2.5-VL-7B + LoRA/PEFT (4-bit) · LangGraph agent · GDAL/rasterio · FastAPI backend · JS mission console · Docker (deployed: GitHub Pages + Render).
- **Methodology:** Query → task classification → input validation (count/modality/CRS/co-registration) → specialist selection from registry → execution → evidence-grounded output + auditable trace:



FEASIBILITY AND VIABILITY

- **Feasibility:** 100% open datasets (BigEarthNet.txt, VRSBench, RSVQA, CDVQA); strong open RS baselines (GeoChat, RemoteCLIP); LoRA trains <1% of parameters on a free T4 GPU; frontend + agentic FastAPI backend already deployed and public.
- **Challenges & risks:** Limited GPU compute · GeoTIFF/multi-band handling · domain gap to Cartosat-2S/RISAT evaluation data · VLM hallucination · orchestration failing live.
- Strategies to overcome:
 - 4-bit quantisation + LoRA adapters (compute); GDAL preprocessing — band selection, CRS checks, tiling (formats)
 - Multisensor S1+S2 training + SAR-style speckle/resolution augmentation (domain transfer)
 - Grounded outputs + confidence thresholds + strict input validator (hallucination)
 - Deterministic routing + full execution trace + graceful fallback to single-image VQA (reliability)

IMPACT AND BENEFITS

- **Target audience:** District officials, disaster-response teams, planners, farmers' bodies and NGOs can question satellite imagery in plain language — no GIS training required.
- **Social:** Democratises ISRO Earth-observation data for non-experts across governance and civil society.
- **Economic:** Hours of manual multi-tool GIS analysis reduced to seconds per query; one platform replaces many single-task systems.
- **Environmental:** Continuous monitoring of forests, water bodies, cropland and urban sprawl; SAR works through clouds, day and night — critical for floods and cyclones.
- **Strategic:** Multiplies the value of Cartosat & RISAT missions; auditable, evidence-grounded AI suitable for official use.

- **Live prototype:** gangadhar017.github.io/Lunar_circle · Source + backend: github.com/Gangadhar017/Lunar_circle
- **Training dataset:** BigEarthNet.txt — [arXiv:2603.29630](https://arxiv.org/abs/2603.29630) · txt.bigearth.net (464,044 co-registered S1+S2 pairs, 9.6M annotations; captions, VQA, referring-expression detection)
- **Evaluation benchmarks:** VRSBench — github.com/lx709/VRSBench · RSVQA — rsvqa.sylvainlobry.com · CDVQA — github.com/YZHJessica/CDVQA
- **Key research:** GeoChat (CVPR 2024) · RemoteCLIP (2024) · RSVQA (IEEE TGRS 2020) · CDVQA (IEEE TGRS 2022) · LoRA (ICLR 2022)
- Final evaluation readiness: prescribed public test splits with normalised scores; multisensor training targets the ISRO/SAC Cartosat-2S + RISAT hidden set.

